

Single core Copper & Aluminum Conductors

XLPE Insulated and PVC Sheathed

Voltage Grade

12 / 20 (24) KV

Description

Soft annealed Stranded circular compacted Copper or Stranded circular compacted Aluminum conductor, semiconducting layer as conductor screen, XLPE insulated semiconducting layer as non metallic insulation screen, Copper tape or wire as metallic insulation screen, and PVC sheathed.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60502
Conductor	Plain copper conductor class 2 or Aluminum conductor class 2
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded or Strippable) to the insulation Metallic part: (by one of the following) • Copper tape is applied helically with a suitable overlap. • Copper Wires helically applied and banded with a Copper tape to achieve electrical contact • Overlapped Copper Tape then Copper Wires are also available (in single core only) Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Outer Sheath	Extruded PVC Compound
Outer Sheath Color	Black (default color), or any other color according to customer request
Maximum Operating Temp.	90° C
Minimum bending Radius	12 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request

Note:-

Flame retardant PVC sheathed or Polyethylene Sheathed are available according to customer request
Longitudinal or radial Water tightness layers are also available according to customer request
the below data are applicable for cables with rated voltage 12.7/22 (24) KV

MEDIUM VOLTAGE CABLES

Product Code	Nominal Cross Sectional Area	Max. Conductor Resistance			Operating Capacitance	Inductance		Current Rating					Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90° C			Flat	Trefoil	Laid in Ground			Laid in Free Air			
			Flat	Trefoil				Flat	Trefoil	Duct	Flat Touched	Trefoil		
														
mm²	Ω/km	Ω/km	Ω/km	μF/km	mH/km	mH/km	A	A	A	A	A	mm	kg/km	
A - Copper Conductors														
PM-CT14-X8-01-UP	50	0.3870	0.4937	0.4937	0.1836	0.4775	0.4313	226	218	189	246	241	26.2	1040
PM-CT15-X8-01-UP	70	0.2680	0.3420	0.3421	0.2087	0.4527	0.4064	278	267	232	308	301	28.2	1285
PM-CT16-X8-01-UP	95	0.1930	0.2466	0.2466	0.2294	0.4335	0.3873	331	316	277	373	365	29.7	1550
PM-CT17-X8-01-UP	120	0.1530	0.1957	0.1958	0.2486	0.4168	0.3706	375	362	317	429	420	31.1	1800
PM-CT18-X8-01-UP	150	0.1240	0.1589	0.1590	0.2692	0.4039	0.3576	415	404	353	484	474	32.6	2165
PM-CT19-X8-01-UP	185	0.0991	0.1274	0.1277	0.2937	0.9190	0.3457	466	456	402	556	545	34.6	2555
PM-CT20-X8-01-UP	240	0.0754	0.0975	0.0979	0.3429	0.3765	0.3303	535	528	468	655	642	36.9	3115
PM-CT21-X8-01-UP	300	0.0601	0.0784	0.0789	0.3547	0.3664	0.3202	594	591	527	748	735	39.3	3745
PM-CT22-X8-01-UP	400	0.0470	0.0623	0.0629	0.3912	0.3559	0.3097	663	665	600	852	842	42.2	4685
PM-CT23-X8-01-UP	500	0.0366	0.0498	0.0507	0.4357	0.3452	0.2990	742	749	682	979	969	45.7	5785
PM-CT24-X8-01-UP	630	0.0283	0.0401	0.0412	0.4909	0.3369	0.2907	833	842	755	1115	1108	49.8	7125
PM-CT25-X8-01-UP	800	0.0221	0.0331	0.0345	0.5500	0.3281	0.2818	910	931	863	1262	1257	54.6	8995
PM-CM26-X8-01-UP	1000	0.0176	0.0251	0.0254	0.6814	0.3296	0.2834	1071	1095	1026	1524	1529	65.4	11955
B - Aluminum Conductors														
PM-AT14-X8-01-UP	50	0.6410	0.8219	0.8219	0.1892	0.4708	0.4246	175	169	146	191	188	26.6	765
PM-AT15-X8-01-UP	70	0.4430	0.5682	0.5682	0.2101	0.4503	0.4041	214	207	179	239	234	28.3	875
PM-AT16-X8-01-UP	95	0.3200	0.4106	0.4106	0.2322	0.4314	0.3851	256	247	216	290	284	29.9	990
PM-AT17-X8-01-UP	120	0.2530	0.3248	0.3248	0.2500	0.4158	0.3696	291	281	246	334	326	31.2	1095
PM-AT18-X8-01-UP	150	0.2060	0.2646	0.2647	0.2733	0.4015	0.3553	323	315	276	377	370	32.9	1290
PM-AT19-X8-01-UP	185	0.1640	0.2109	0.2111	0.2978	0.3899	0.3437	365	358	316	435	428	34.9	1460
PM-AT20-X8-01-UP	240	0.1250	0.1612	0.1614	0.3249	0.3765	0.3303	421	414	366	514	503	36.9	1660
PM-AT21-X8-01-UP	300	0.1000	0.1293	0.1296	0.3547	0.3664	0.3202	466	465	414	590	577	39.3	1895
PM-AT22-X8-01-UP	400	0.0778	0.1012	0.1016	0.3912	0.3559	0.3097	533	530	477	682	670	42.2	2320
PM-AT23-X8-01-UP	500	0.0605	0.0796	0.0802	0.4357	0.3452	0.2990	602	605	550	779	781	45.7	2730
PM-AT24-X8-01-UP	630	0.0469	0.0627	0.0635	0.4922	0.3367	0.2905	686	686	621	920	907	49.9	3255
PM-AT25-X8-01-UP	800	0.0367	0.0505	0.0515	0.5500	0.3281	0.2818	775	775	719	1055	1042	54.6	3930
PM-AM26-X8-01-UP	1000	0.0291	0.0390	0.0392	0.6814	0.3296	0.2834	911	920	862	1290	1282	65.4	5325

The above data is approximate and subject to manufacturing tolerances